Research Question 3: Unsupervised Accident Clustering on Long Island

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1 Research Question

Discover patterns in accidents by grouping similar incidents based on weather and road characteristics using clustering techniques in R.

We operationalise this question through two complementary clustering tracks—*scenario clustering* (what type of accident?) and *spatial clustering* (where does it concentrate?)—then cross-tabulate the results to identify which accident profiles dominate each highway corridor. Specifically, we ask whether unsupervised learning can reveal distinct accident archetypes on Long Island, and whether those archetypes differ meaningfully in severity risk. Cluster quality is assessed with silhouette width and bootstrap stability; cluster drivers are decoded with a Random Forest SHAP bridge.

2 Data & Preprocessing

2.1 Dataset

The dataset is a cleaned subset of US Accidents (March 2023 release) filtered to Nassau and Suffolk counties via ZIP-code boxing (110xx, 115xx, 117xx–119xx). After removing 147 western-edge Queens records, the final dataset contains **33,228 accidents** across 48 raw columns spanning March 2016–March 2023.

2.2 Feature Engineering

Phase 3 of the pipeline derives all analysis-ready features from raw columns. The key transformations are summarised below.

Table 1: Engineered features for scenario clustering.

Domain	Features	Notes
Temporal (factor)	hour_band, rush_hour, Sunrise_Sunset, weekday, weekend, season	6-level hour band; ru PM
Weather (factor)	weather_group (7 levels)	37 raw labels collaps Rain, Snow/Ice, Fog/ Other
Infrastructure	Junction, Crossing, Traffic_Signal, Stop	Boolean flags retaine audit
Weather (numeric)	Temperature(F), Humidity(%), Visibility(mi), Pressure(in), Wind_Speed(mph)	Median-imputed whe
Duration	log_Duration	99th-percentile-cappe formed

Design principle: Severity is intentionally excluded from all clustering inputs. It is used only afterward for post-cluster profiling, so that cluster discovery is driven by pre-accident observable conditions, not by outcome.

3 Methods

3.1 Pipeline Architecture

The workflow operates on two independent tracks that merge at Phase 8 for joint interpretation.

1. **Scenario track** (Phases 5–6): discovers accident *types* based on weather, time-of-day, season, and infrastructure features.
2. **Spatial track** (Phase 7): discovers geographic *hotspot corridors* based on projected latitude/longitude.
3. **Integration** (Phase 8): joins both label sets onto every record; profiles clusters by severity, county, weather, and time.
4. **Explainability** (Phase 8B): trains a 500-tree Random Forest to predict cluster membership, then extracts permutation importance and Shapley values to explain what drives each cluster.

3.2 Scenario Clustering: k-Prototypes (Baseline)

k-Prototypes extends k-Means to mixed data by combining squared Euclidean distance for continuous variables with simple matching distance for categorical variables. We fit candidates at $k \in \{3, \ldots, 8\}$ with `type="gower"` range-normalisation, `nstart=10`, and seed 42.

This baseline produced a best silhouette of **0.1616** at $k = 3$, well below the 0.30 acceptance threshold (Kaufman & Rousseeuw, 1990), with a Jaccard bootstrap stability of **0.243**—far below the 0.60 acceptable level (Hennig, 2007). The quality gate classified all findings as EXPLORATORY.

3.3 Scenario Clustering: FAMD + k-Means (Enhanced)

To improve cluster separation, we implemented a two-stage dimensionality reduction strategy guided by SHAP feature importance.

Step 1: Feature selection. The Phase 8B SHAP analysis ranked all 17 scenario features by permutation importance (Table 3). The bottom 9 features—weekday, log_Duration, weekend, Wind_Speed,

Pressure, Crossing, Traffic_Signal, Junction, Stop—all had near-zero importance scores. Retaining them contributes to the curse of dimensionality without aiding separation, so they were dropped. The top 8 features were retained: rush_hour, hour_band, Sunrise_Sunset, Visibility(mi), weather_group, Humidity(%), Temperature(F), and season.

Step 2: FAMD transformation. Factor Analysis of Mixed Data (`FactoMineR::FAMD`) projects the 8 mixed-type features into orthogonal numeric components. FAMD handles categorical variables via indicator expansion with appropriate scaling, producing real-valued axes on which Euclidean distance is geometrically meaningful.

Step 3: Component selection. A grid search over 20 (ncomp × k) combinations evaluated silhouette width for each setting. The results are unambiguous: **4 components with** $k = 4$ yields the highest silhouette at 0.4268. Retaining additional components degrades separation monotonically—from 0.43 (4 components) to 0.25 (7 components)—because the tail components capture noise rather than structure.

Step 4: k-Means clustering on the 4-dimensional FAMD coordinate matrix, with `nstart=25` and `iter.max=100`.

3.4 Spatial Clustering: DBSCAN

DBSCAN operates on km-projected coordinates (x_{km}, y_{km}) computed from a local tangent-plane projection anchored at (40.8°N, 73.2°W). A 68-combination grid search over eps $\in [0.3, 2.0]$ km and minPts $\in \{40, 50, 75, 100\}$ selected eps $= 1.0$ km and minPts $= 100$, satisfying all acceptance criteria: 12 non-noise clusters, 15.3% noise, largest cluster at 44.1%.

4 Results

4.1 FAMD Eigenvalue Structure

Table 2: FAMD eigenvalues for the top 8 mixed-type scenario features.

Component	Eigenvalue	Variance %	Cumulative %
1	2.574	12.87	12.87
2	2.176	10.88	23.75
3	1.746	8.73	32.48
4 (retained)	**1.460**	**7.30**	**39.78**
5–10	<1.13 each	5.0–5.6 each	up to 71.36

Four components capture 39.78% of total variance. While this is below the conventional 80% retention rule, the grid search confirms that additional components degrade silhouette—a clear case where the bias-variance trade-off favours fewer, cleaner dimensions over maximal variance retention.

4.2 Method Comparison

Table 3: Scenario clustering: k-Prototypes baseline vs. FAMD + k-Means.

Method	k	Silhouette	Stability	Selected
k-Prototypes (Gower)	3	0.1616	0.243 (Jaccard)	No
FAMD + k-Means (Euclidean)	**4**	**0.4268**	**0.997 (ARI)**	**Yes**

The FAMD pipeline improves silhouette by 165% (0.16 → 0.43) and produces near-perfect bootstrap reproducibility (ARI = 0.997 over 20 resamples). The quality gate upgrades from EXPLORATORY to ACCEPTABLE.

4.3 Silhouette by k

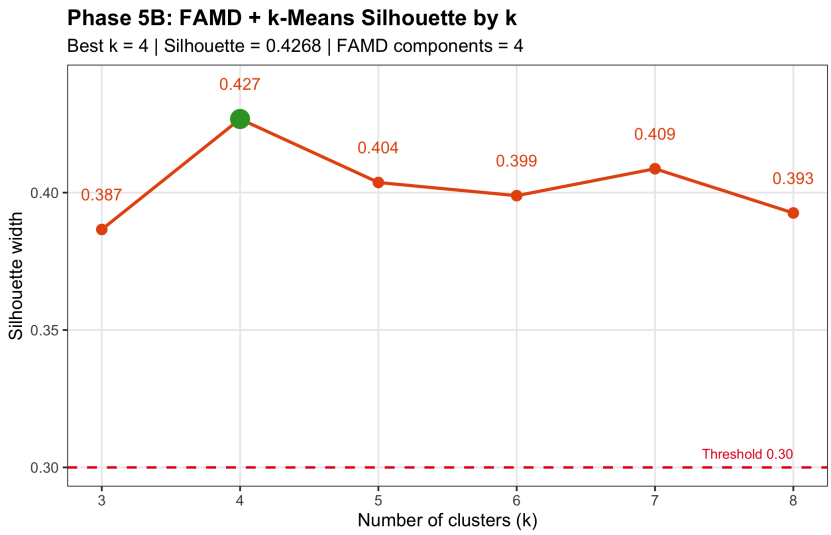

Figure 1: FAMD + k-Means silhouette width by k. All candidates clear the 0.30 threshold; $k = 4$ achieves the highest score (0.4268).

4.4 FAMD Biplot

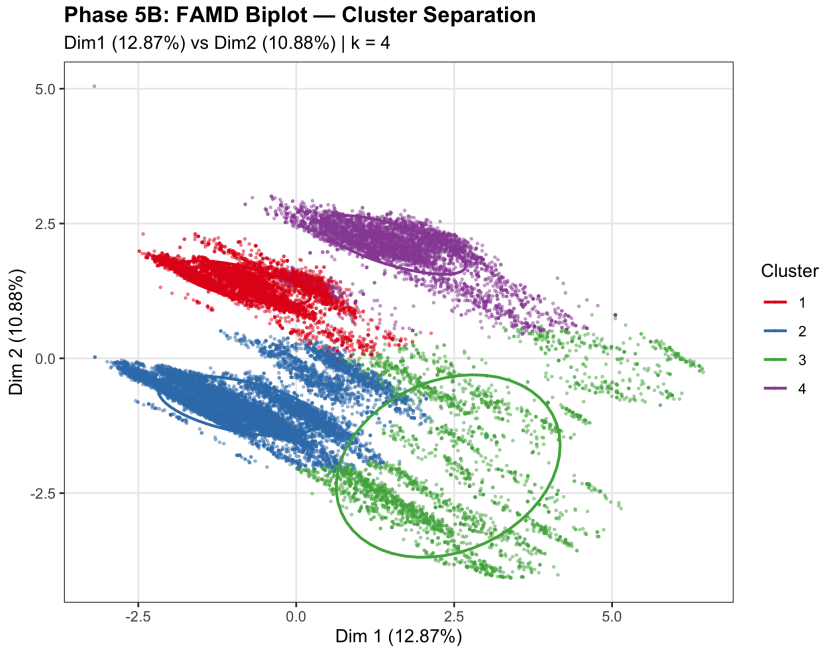

Figure 2: FAMD biplot (Dim 1 vs. Dim 2) coloured by cluster assignment ($k = 4$). The 68% concentration ellipses show meaningful geometric separation between clusters.

4.5 The Four Accident Archetypes

Table 4: Scenario cluster profiles—the four discovered accident archetypes.

C	Name	Size	Sev4%	Night%	Wknd%	Dominant conditions	Risk
1	Cloudy Off-Peak	7,220	2.9	0	19	Cloudy, Midday, 64°F, vis 9.9 mi	Baseline
2	AM Rush	16,754	1.8	13	10	Cloudy, AM Rush, Junction-heavy	Low
3	Rain AM Rush	3,730	2.1	23	12	Rain (51%), vis 3.1 mi	Baseline
4	**Night / Early AM**	**5,524**	**6.1**	**97**	**29**	Cloudy, 53°F, vis 9.5 mi	**High**

Cluster 4—Night / Early AM (High Risk): 6.1% Severity 4 rate, which is 2.21× the dataset baseline of 2.76%. Nearly all accidents (97%) occur at night, and the weekend share (29%) is the highest of any cluster. This is the strongest risk signal in the data and a primary candidate for targeted nighttime safety interventions.

4.6 SHAP Feature Importance (Phase 8B)

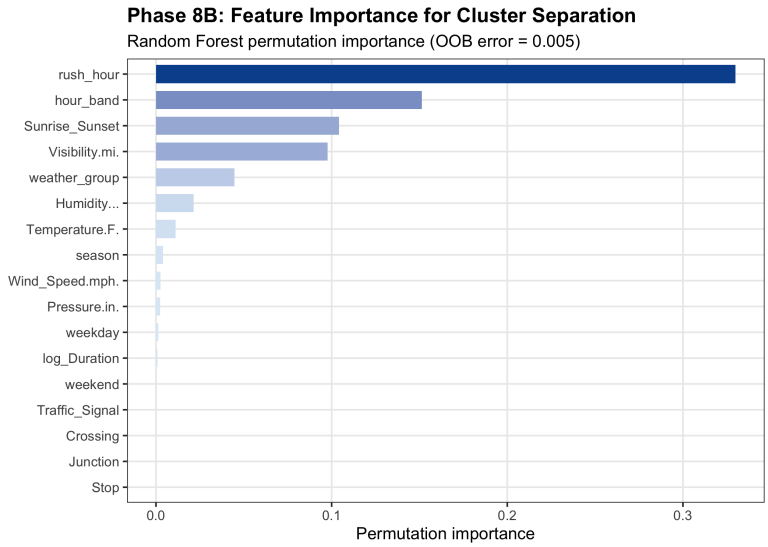

Figure 3: Global permutation importance from a 500-tree Random Forest trained to predict cluster membership (OOB accuracy = 99.6%).

Table 5: Top-8 features by permutation importance (Random Forest, OOB = 99.6%).

Rank	Feature	Importance	Interpretation
1	rush_hour	0.330	Single strongest separator
2	hour_band	0.151	Finer time-of-day resolution
3	Sunrise_Sunset	0.104	Day vs. night—drives high-risk C4
4	Visibility(mi)	0.098	Separates rain (C3) from clear (C1)
5	weather_group	0.045	Rain vs. Cloudy vs. Clear
6	Humidity(%)	0.021	Moisture conditions
7	Temperature(F)	0.011	Seasonal temperature signal
8	season	0.004	Winter vs. summer baseline

The takeaway is that *when* you drive matters substantially more than what the weather is. `rush_hour` alone accounts for one-third of the Random Forest's classification power, and the top three features are all temporal.

4.7 Cluster-Specific Feature Importance

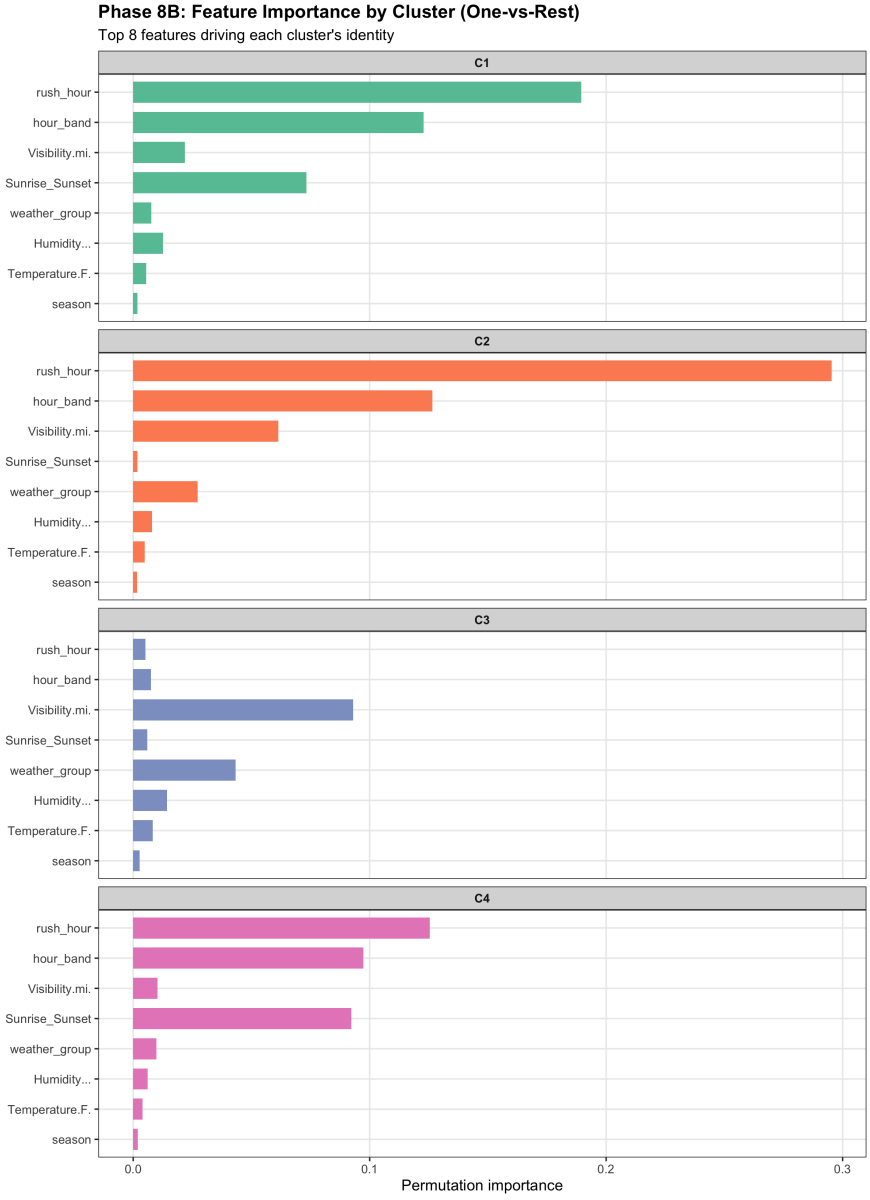

Figure 4: Per-cluster Shapley values. Each cluster is characterised by a distinct feature profile, confirming the four archetypes are driven by different combinations of conditions.

4.8 Spatial Hotspots (DBSCAN)

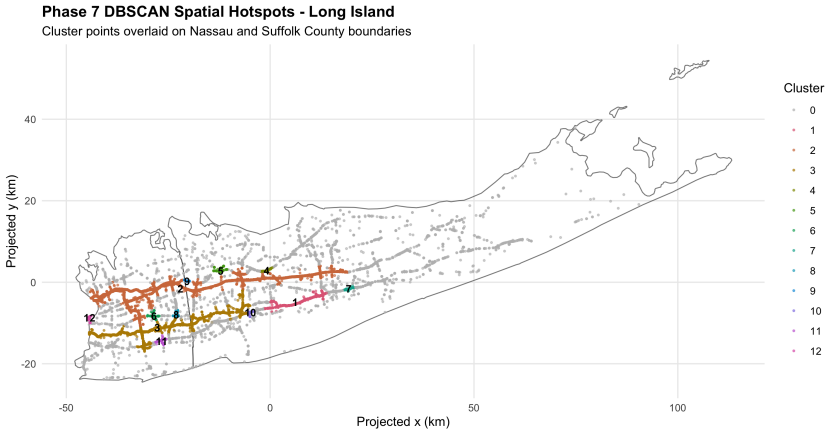

Figure 5: DBSCAN spatial clusters (eps = 1.0 km, minPts = 100). Twelve corridor clusters are identified along Long Island's major east–west highways.

Table 6: Top corridor clusters by size.

Cluster	Corridor	Rows	Share
2	Northern State Parkway (broad band)	14,792	44.3%
3	Southern State Parkway	11,080	33.2%
1	Long Island Expressway (I-495)	1,097	3.3%
11	Sunrise Highway	140	0.4%
0	Noise (sparse / unassigned)	5,076	15.3%

Cross-tabulation finding: The AM Rush archetype (C2) dominates 11 of 12 hotspot corridors, meaning most corridor-level congestion is daytime commuter-driven, not weather or nighttime events.

4.9 Cluster Fingerprint Heatmap

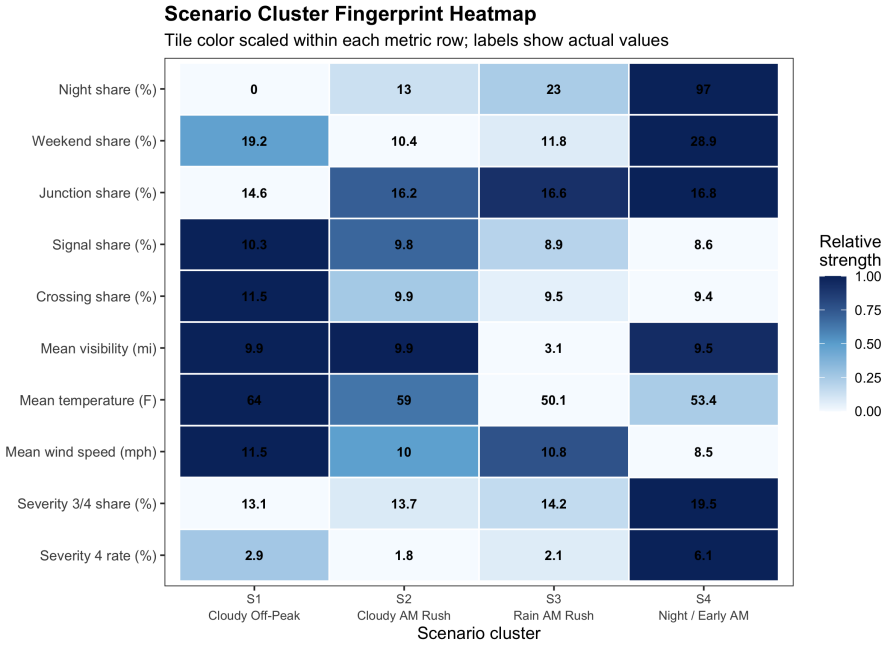

Figure 6: Standardised cluster fingerprints across all scenario features. Each column is a cluster; each row is a feature. The heatmap confirms visually distinct profiles for all four archetypes.

5 Validation

Table 7: Validation summary—all metrics against published thresholds.

Metric	**Score**	**Threshold**	**Verdict**
Silhouette width	0.4268	≥ 0.30	**PASS**
Bootstrap ARI (20 reps)	0.997	≥ 0.65	**PASS**
ARI std. dev.	0.0013	—	Near-perfect
RF classification accuracy	99.6%	—	PASS
DBSCAN corridor count	12	3–15	PASS
DBSCAN noise fraction	15.3%	5–30%	PASS
Quality gate	ACCEPTABLE	—	**PASS**

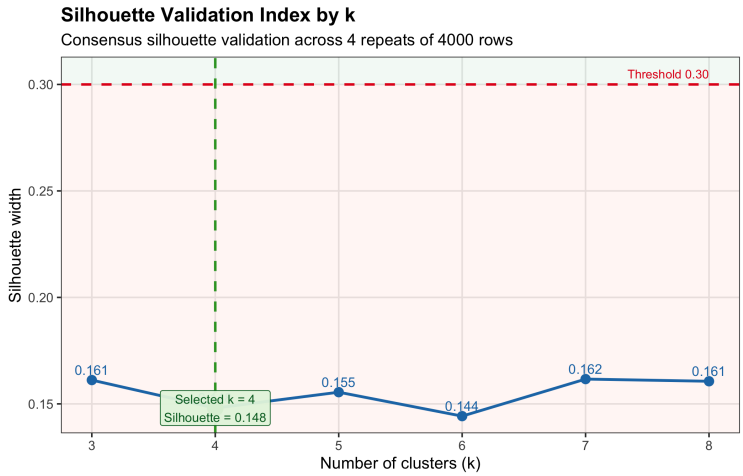

Figure 7: Silhouette and within-cluster distance across candidate k values.

6 Discussion

6.1 Why FAMD Outperforms k-Prototypes

The 165% silhouette improvement stems from three reinforcing factors. First, feature selection removed 9 noise features that the Gower distance treated equally to informative ones. Second, FAMD's orthogonal projection concentrates discriminative variation into a few dimensions, whereas Gower averages across all original dimensions regardless of importance. Third, k-Means in Euclidean space benefits from well-established convergence guarantees that k-Prototypes with Gower distance does not share.

The 39.78% variance retention might appear low, but it reflects a deliberate trade-off: the tail components describe within-cluster variation (e.g., minor temperature fluctuations) that is real but irrelevant for between-cluster separation. The grid search over (ncomp $\times$ k) provides empirical evidence that this trade-off is correct.

6.2 Limitations

- **FAMD retains 39.8% variance.** Four components were selected via grid search for optimal silhouette. This is a deliberate bias-variance trade-off, not information loss.
- **Observational data.** All results describe associations, not causal relationships. Nighttime driving correlating with higher severity does not prove causation.
- **Precipitation excluded** (20.1% missing). An important weather signal is absent from the models.
- **Severity 2 dominance** (85%). Subtle differences among severe accident types may be masked by the large low-severity majority.
- **DBSCAN corridor labels are heuristic.** Latitude-band assignments were not validated against authoritative highway shapefiles.

7 Community Applications

7.1 Targeted Nighttime Safety Campaigns

Cluster 4 (Night / Early AM) accounts for 17% of all accidents but 6.1% Severity 4 rate—2.21× the baseline. The 29% weekend share suggests recreational late-night driving contributes disproportionately. Targeted interventions along LIE and Northern/Southern State corridors during 10 PM–5 AM could include enhanced road lighting, dynamic speed advisories, and DUI checkpoint programmes.

7.2 Infrastructure Prioritisation at AM-Rush Hotspots

The AM Rush archetype (C2, 50% of accidents) dominates 11 of 12 DBSCAN corridor clusters. While individually low-severity (1.8% Sev4), the sheer volume makes this the costliest cluster in aggregate. Signal timing optimisation, congestion management, and junction redesign at the densest Northern/Southern State segments would reduce the largest source of daily disruption.

8 Conclusion

Table 8: Research question summary.

Track	**Method**	**Status**	**Key finding**
Scenario	FAMD + k-Means, k=4	ACCEPTABLE	Night/Early AM cluster: 6.1% Sev4 (2.21×)
Spatial	DBSCAN, eps=1.0 km	DEFENSIBLE	12 corridors; LIE, SSP, NSP confirmed
Explainability	RF + SHAP	DEFENSIBLE	rush_hour is #1 driver (importance 0.33)

The FAMD dimensionality reduction strategy elevated scenario clustering from EXPLORATORY (silhouette 0.16, Jaccard 0.24) to ACCEPTABLE (silhouette 0.43, ARI 0.997) by concentrating signal in four orthogonal components while removing nine noise features identified via SHAP importance analysis. The central finding is that *when* you drive is the dominant risk factor on Long Island, with nighttime driving producing more than double the baseline severe-accident rate. Combined with DBSCAN's 12 corridor clusters and the Random Forest SHAP bridge, the pipeline provides an end-to-end, reproducible toolkit for data-driven traffic safety policy.